

David Shu-Yu Wu

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EDUCATION

National Taiwan University

B.S. in Computer Science and Information Engineering; Minor in Atmospheric Sciences

Taipei, Taiwan

Sep. 2023 – Present

PROFILE

Senior CS undergraduate at National Taiwan University focused on cryptographic engineering, post-quantum cryptography, and implementation security. Research experience includes high-performance NTT-based big-integer arithmetic on AArch64, formal verification of optimized cryptographic kernels, and side-channel trace generation and evaluation using diffusion models. Broader academic interests include theoretical computer science, algorithms, and probabilistic methods.

RESEARCH EXPERIENCE

Fast Crypto Lab

Jul. 2025 – Present

Research Intern, Academia Sinica, supervised by Bo-Yin Yang

Taipei, Taiwan

Multiplying Not-So-Big Integers with FFTs

Finding the Crossover Point on Large Modern OoO ARM CPUs

Accepted Poster, CHES 2026

- Developed constant-time NEON-optimized NTT-based large-integer multiplication on AArch64, including fused NTT kernels, Barrett/Montgomery arithmetic, Good's trick, CRT reconstruction, and chunking/dechunking.
- Integrated the NTT-based multiplier into OpenSSL RSA-8192/10240, improving verification, encryption, and KEM encapsulation by 36.0%–44.7%.
- Formally verified optimized assembly kernels with CryptoLine for algebraic correctness, range safety and checked equivalence after Slothy scheduling. Used HOL Light to compose higher-level multiplication specifications.
- Continued optimizing the multiplication pipeline with more aggressive lazy reduction, fused reconstruction, and dependency-chain elimination, further reducing the practical crossover point against GMP.

Nanyang Technological University

Jul. 2026 – Present

Global Connect Fellow (Jul.–Aug. 2026); Remote Collaborator

Singapore / Remote

Generative AI for High-Fidelity Side-Channel Trace Simulation

Manuscript in Preparation

- Selected for the NTU Global Connect Fellowship, a competitive international summer research program with an acceptance rate below 2%; supervised by Gwee Bah Hwee and Juncheng Chen.
- Built controlled synthetic side-channel datasets and conditional DDPM experiments to study how attack-relevant leakage emerges across the 400-stage reverse diffusion process.
- Evaluated intermediate synthetic traces using Template-attack NTGE and CPA correct-key peak/margin, together with attack-independent waveform and distribution fidelity metrics including RMSE, NCC, DTW, Wasserstein-1, and Energy distance.
- Observed in controlled synthetic traces that attackability can peak before full waveform fidelity is recovered, revealing a temporal separation between cryptanalytic utility and trace fidelity.

COURSE PROJECTS

Cryptographic Engineering: ML-KEM & X25519 on Cortex-M4 | C, ARMv7E-M ASM

Fall 2025

- Optimized constant-time X25519 (ECDH25519) on Cortex-M4, including variable-base scalar multiplication and fixed-base window method (constant-time table scan + conditional move).
- Reworked field arithmetic modulo $2^{255} - 19$: implemented mixed-radix (25/26-bit limbs) multiplication with product-scanning, and specialized squaring to eliminate redundant cross terms; translated hotspots to ARMv7E-M assembly to reduce cycle counts.
- Optimized ML-KEM polynomial arithmetic: implemented NTT/InvNTT and base multiplication in assembly with layer fusing and controlled unrolling; leveraged ARMv7E-M DSP packed 16-bit operations for poly add/sub under strict NTT code-size budgets.
- Achieved up to **94%** cycle-count reduction for X25519 fixed-base scalar multiplication and **62%** reduction for ML-KEM `poly_ntt` vs. course baseline under the same benchmarking setup.

Fuzzy Counting with Probabilistic Data Structures | C++, Python

Spring 2025

- Implemented typo-tolerant fuzzy matchers using Bloom filter segmentation, Cuckoo filters, HyperLogLog, and HyperLogLog-LDP.
- Evaluated accuracy–space tradeoffs over synthetic typo datasets under different sketch parameters and segmentation strategies, exposing the implementation as a reusable Python library.

- Extended xv6 with user-level threads, cooperative context switching, and a SLAB allocator with freelist management and partial/full slab classification.
- Implemented and evaluated CPU schedulers (HRRN, P-RR, DM, EDF-CBS) and simulated RAID-1 fault-tolerant file operations in a Docker-based development environment.

COURSES

Relevant Coursework: Cryptography and Network Security, Post-Quantum Cryptography, Cryptographic Engineering, Cryptanalysis, Algorithm Design and Analysis, Advanced Data Structures, Probability, Discrete Mathematics, Automata and Formal Languages, Information Theory and Coding Techniques, Operating Systems, Computer Architecture, Data Structures and Algorithms, System Programming, Linear Algebra

TECHNICAL SKILLS

Languages: C, C++, Python, AArch64 assembly, ARMv7E-M assembly

Cryptography / Verification: ARM NEON intrinsics, CryptoLine, SLOTHY

Tools & Platforms: Git, Docker, Linux, Raspberry Pi 5, VS Code